SynapticLifetimes: summary of changes

Branch `modernize-and-cleanup` (24 commits ahead of `master`), October 2026

The code was ported to current dependencies, all simulation mechanisms were reviewed one by one, unused mechanisms and code were removed, several bugs that changed the results were fixed, and the memory footprint of the network was reduced so that larger networks can be simulated. Each step was verified by comparing the simulation output with the previous version (bit-identical wherever the change was not meant to alter results).

1. Environment

- Python 3.12 and upstream Brian2 2.10 (instead of a patched Python 3.8 environment); pinned in `network/environment.yml` (conda environment `synlife`).
- Removed the GPU path (brian2cuda), the vendored Brian2 copy and the Elephant tarball; neo, elephant and quantities are no longer needed.

2. Bugs fixed that change the results

- **STDP timing:** the plastic synapses ran on a 100 ms clock, so STDP and iSTDP used spike times rounded to 100 ms (effectively a 100 ms coincidence rule). They now run on the default clock; the weight noise is applied every `synEE_mod_dt` with its exact solution.
- **Distance-dependent insertion:** `p_distance` was assigned the full $N_e \times N_e$ matrix, so insertion probabilities were attached to the wrong synapses.
- **Distance-dependent delays:** assigned to the wrong synapses, with a constant 0.1 s standard deviation (half of them clipped to 0). Now distance/velocity with a relative SD.
- Inhibitory neurons used the excitatory refractory period; `generate_connections` excluded the wrong target neuron.
- iSTDP amplitudes were only set with distance-dependent connectivity; with `stdp_active=0` all EE weights were set to zero; Poisson input onto inhibitory neurons used the excitatory weight; biexp conductances used the wrong mode for I→I.
- Growth homeostasis: `adjust_rate` aimed at a hard-coded 10% instead of `p_ee`; `constant_count` inserted a burst of synapses at $t = 0$.
- Analysis: lifetime extraction dropped the last synapse; `t5_peaks` used a hard-coded $N_e = 400$ and a wrong weight-matrix layout; several post-processing crashes.

3. Model changes (decided during the review)

- STDP and iSTDP use all-to-all spike interactions (instead of nearest-spike).
- Synaptic weight noise: only multiplicative noise, on EE and EI synapses, with separate strengths `syn_sigma_EE`, `syn_sigma_EI`.
- Distance-dependent connectivity: each pair is connected independently with probability $\min(1, c\,p(d))$ (Gaussian, clipped if needed), i.e. the same profile as the insertion of new synapses. `half_width` is renamed `sigma_dd`; new `sigma_dd_inh`.
- All plasticity (including the adaptive normalization target, the growth homeostasis and the weight noise) is frozen in T5.
- Separate excitatory and inhibitory neuron models.

4. Removed

- Mechanisms not used by any configuration: Kesten, decay and additive noise, intrinsic plasticity, spike-rate adaptation, strong membrane and Cramer noise, triplet/Jedlicka STDP, short-term depression, alpha conductances, symmetric iSTDP, inhibitory structural plasticity, loading weights from file, alternative normalization modes, lognormal initial weights.
- Unused analysis scripts (parameter sweeps, duplicates, notebooks, multi-run framework).
- Redundant recordings: population rates (now computed from the spikes, identical), cross-correlations, resampled STDP/scaling files, threshold traces.

5. Memory and speed

- No arrays over all $N_e(N_e - 1)$ potential EE pairs are built in Python: the pairs, the distance-dependent probabilities and the delays are generated in C++, and the connectivity is computed block-wise.
- The distance factor `p_distance` is computed on the fly instead of stored per pair; active synapses are counted per neuron.
- Weight snapshots store only the active synapses (read with `analysis.utils.WeightSnapshots`); about 10$\times$ smaller files.
- The weight-noise operation is switched off in T5 (it drew a random number per potential pair every 100 ms).

	Python (peak)	C++ simulation (peak)	wall time (36 s simulated)
$N_e = 1600$, before	2173 MB	1596 MB	178 s
$N_e = 1600$, now	1072 MB	341 MB	155 s
$N_e = 3200$, now	1636 MB	1059 MB	546 s

The weight noise during T1–T4 is now the largest cost (≈30% of the simulation time), because it acts on every potential pair.

6. Things to be aware of

- **Earlier results must be re-run:** the STDP timing fix and the other bug fixes change all plastic configurations.
- Random realizations differ from earlier runs with the same seed (delays drawn in C++, noise off in T5, independent connectivity).
- New output format of `synee_a.p`, `synei_a.p`, `syneedynrec.p`, `syneidynrec.p`; the final snapshot carries the actual end time of the run. Synaptic delays are no longer stored in the pypet results.
- In small test networks the T5 firing rate is very sensitive to the random realization (e.g. 7 vs. 67 Hz for the same seed); compare code versions with several seeds and perturbation controls.
- All network tests pass; three analysis unit tests need fixture files that are not in the repository.